

Distributed Systems

Lecture10: Virtualization

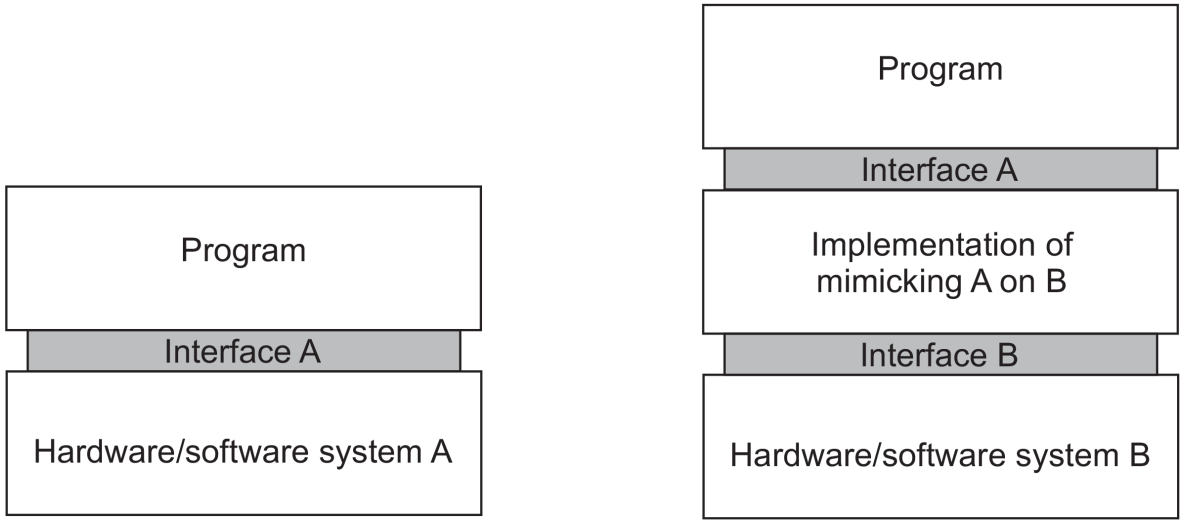
Mohammad Reza Shakournia

shakournia@ut.ac.ir

School of Electrical and Computer Engineering,
University of Tehran,
Tehran, Iran

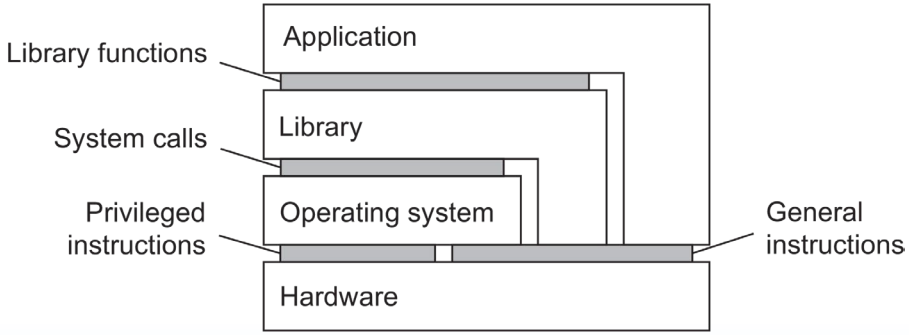
Virtualization

- Virtualization is important:
  - Hardware changes faster than software
  - Ease of portability and code migration
  - Isolation of failing or attacked components

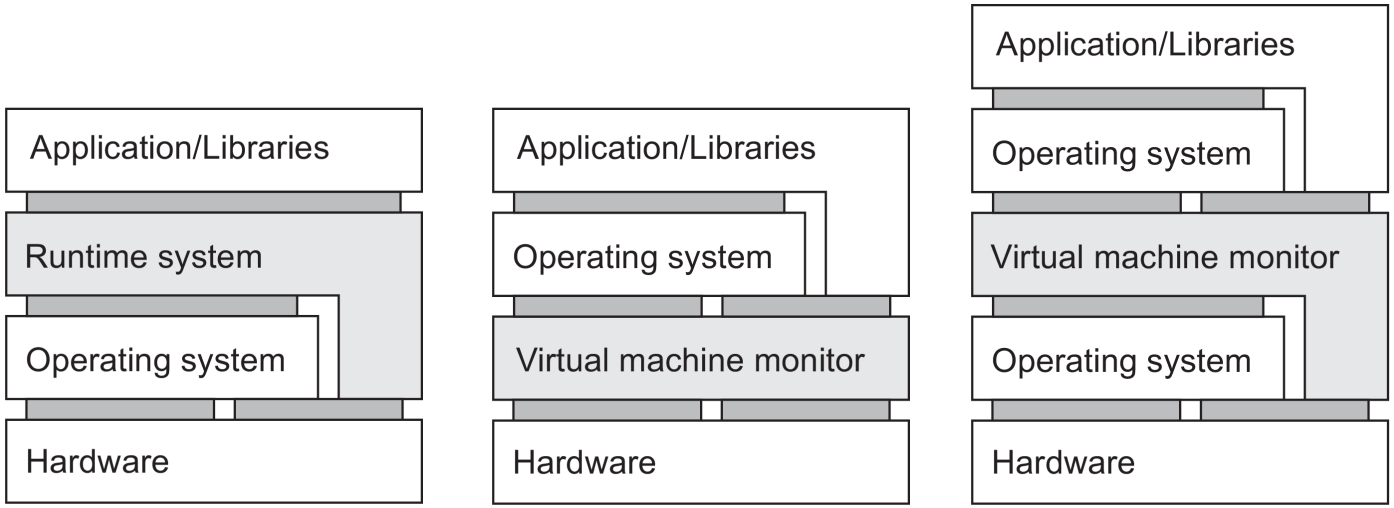


Mimicking Interfaces

- Four types of interfaces at three different levels:
  1. Instruction set architecture: the set of machine instructions, with two subsets:
     - **Privileged instructions:** allowed to be executed only by the operating system.
     - **General instructions:** can be executed by any program.
  2. System calls as offered by an operating system.
  3. Library calls, known as an application programming interface (API)



Virtualization Types



(a) Process VM (b) Native VMM (c) Hosted VMM

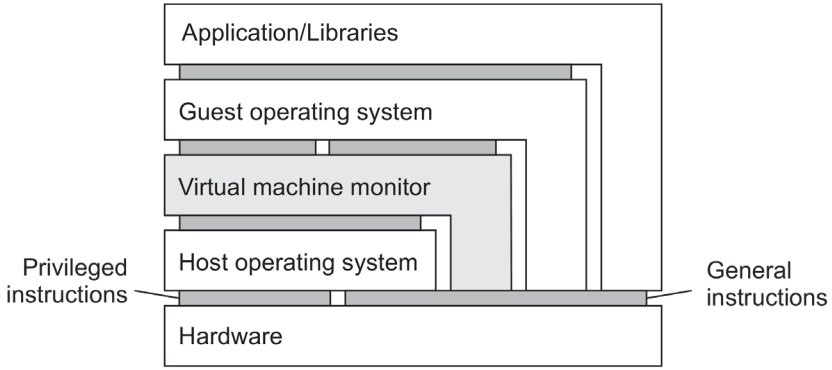
(a) Separate set of instructions, an interpreter/emulator, running a top an OS.

(b) Low-level instructions, along with bare-bones minimal operating system

(c) Low-level instructions, but delegating most work to a full-fledged OS.

Zooming Into VMs: Performance

- **Refining the organization**



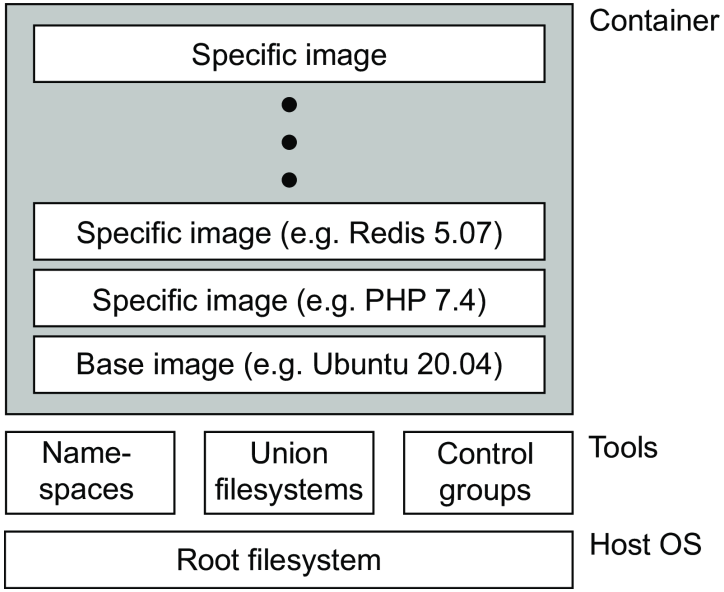
- Privileged instruction: if and only if executed in user mode, it causes a trap to the operating system.
- Nonpriviliged instruction: the rest.

- **Special instructions:**
  - Control-sensitive instruction: may affect configuration of a machine (e.g., one affecting relocation register or interrupt table).
  - Behavior-sensitive instruction: effect is partially determined by context (e.g., POPF sets an interrupt-enabled flag, but only in system mode).

Condition for Virtualization

- **Necessary condition:**
  - For any conventional computer, a virtual machine monitor may be constructed if the set of sensitive instructions for that computer is a subset of the set of privileged instructions.
- **Problem**: condition is not always satisfied!
  - There may be sensitive instructions that are executed in user mode without causing a trap to the operating system.
- **Solutions**
  - Emulate all instructions.
  - Wrap nonprivileged sensitive instructions to divert control to VMM.
  - Paravirtualization: modify guest OS, either by preventing nonprivileged sensitive instructions, or making them nonsensitive (i.e., changing the context).

Containers

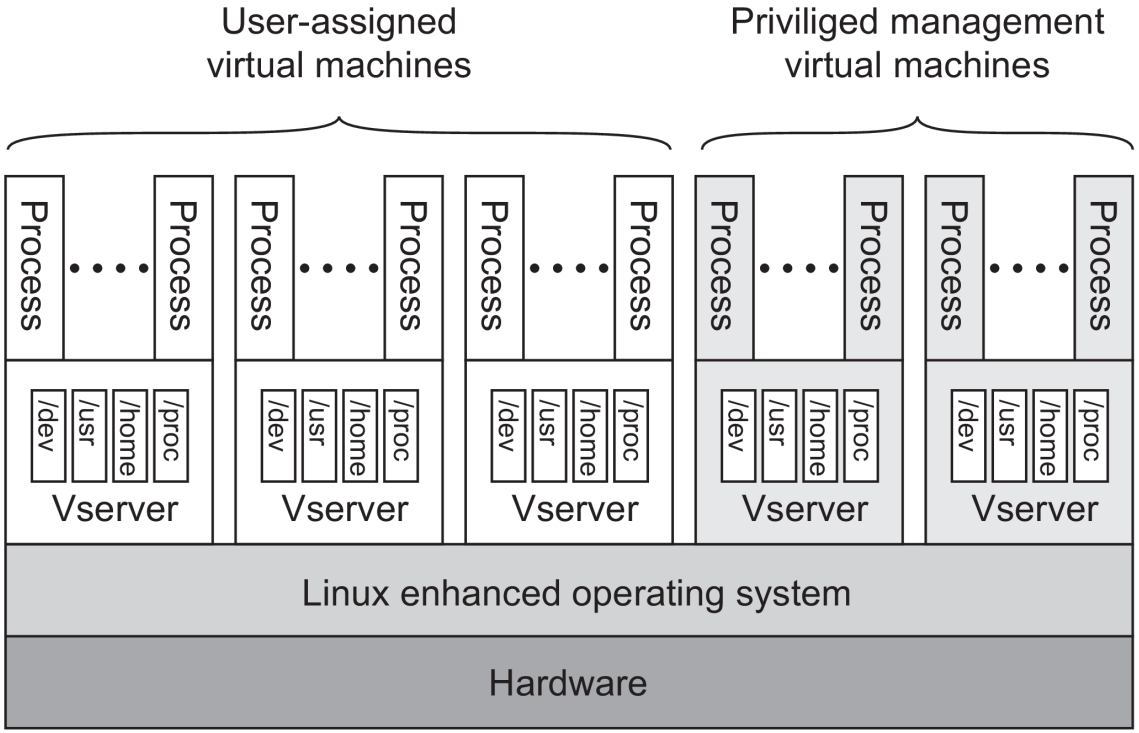


- **Namespaces:** a collection of processes in a container is given their own view of identifiers.
- **Union file system:** combine several file systems into a layered fashion with only the highest layer allowing for write operations (and the one being part of a container).
- **Control groups:** resource restrictions can be imposed upon a collection of processes.

Example: PlanetLab

- Different organizations contribute machines, which they subsequently share for various experiments.
- **Problem:**
  - We need to ensure that different distributed applications do not get into each other's way ⇒ virtualization

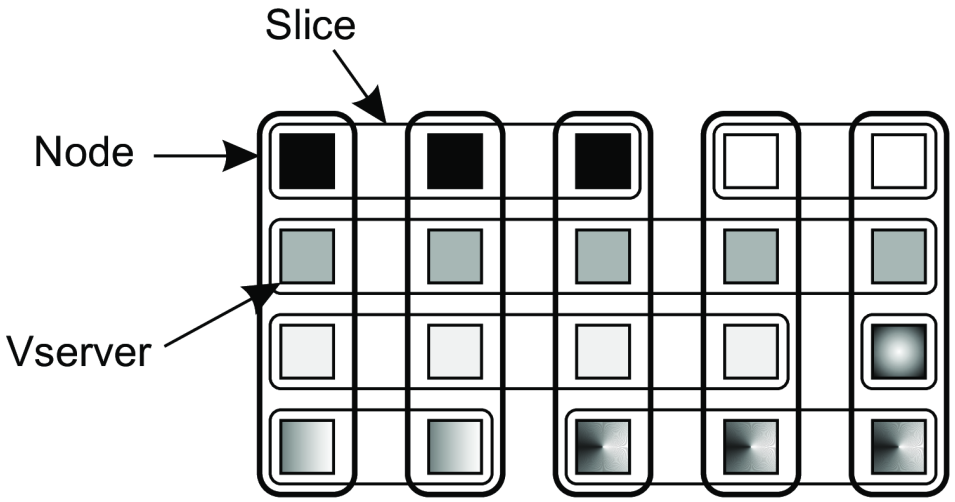
PlanetLab Basic Organization



- **Vserver:** Independent and protected environment with its own libraries, server versions, and so on. Distributed applications are assigned a collection of vservers distributed across multiple machines

PlanetLab Vservers and Slices

- Each Vserver operates in its own environment (cf. chroot).
- Linux enhancements include proper adjustment of process IDs (e.g., init having ID 0).
- Two processes in different Vservers may have same user ID, but does not imply the same user.



VMs and Cloud Computing

- Three types of cloud services:
  - Infrastructure-as-a-Service covering the basic infrastructure.
  - Platform-as-a-Service covering system-level services.
  - Software-as-a-Service containing actual applications.
- **IaaS:** Instead of renting out a physical machine, a cloud provider will rent out a VM (or VMM) that may be sharing a physical machine with other customers ⇒ almost complete isolation between customers (although performance isolation may not be reached).

Wrap-Up + What's Next

In the next lecture, we will discuss the anatomy of clients and servers, respectively including:

- Network User Interface,
- Virtual Desktop Environment,
- Client-side Software for Distributed Transparency.